

Sequence and Series:

Sequences:

$$\left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, \dots\right)$$

$$(2, 4, 6, 8, 10, \dots, 2n, \dots)$$

$$(2, 4, 8, 16, \dots, 2^n, \dots)$$

$$\left(\frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2^n}, \dots\right)$$

$$(-1, 1, -1, 1, \dots, (-1)^n, \dots)$$

$$\left(0.3, 0.33, 0.333, \dots, \underbrace{0.33\dots3}_{(n \text{ times})}, \dots\right)$$

Definition: A sequence (of points) in $\mathbb{R}$ is a function from the set $\mathbb{N}$ of natural numbers to the set $\mathbb{R}$ of real numbers.

Note: A sequence therefore assigns to each natural number n , a uniquely determined real number x_n (or $x(n)$ or $f(n)$)

Notation:

$(x_n)_{n \in \mathbb{N}}$ or (x_n)

(The terms of the sequence)

[Ordering induced by
the natural order of $\mathbb{N}$]

Fibonacci Sequence: $(x_n)_{n \in \mathbb{N}}$

where $x_1 = 1$

$x_2 = 1$

and $x_{n+1} = x_{n-1} + x_n$ ($\forall n \in \mathbb{N}, n \geq 2$)

So, the sequence is $(1, 1, 2, 3, 5, 8, 13, 21, \dots)$

Definition (limit of a sequence):

A sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ is said to converge
to $x \in \mathbb{R}$ if

for every $\epsilon > 0$, there exists a natural
number $K = K(\epsilon)$ (depends on ϵ) such that

for all $n \geq K(\epsilon)$, $|x_n - x| < \epsilon$

The point x is called the "limit" of the
sequence (x_n) and we write

$$\lim_{n \rightarrow \infty} (x_n) = x$$

or $x_n \longrightarrow x$

Note: If $(x_1, x_2, \dots, x_n, \dots)$ is a sequence in $\mathbb{R}$ and if $m \in \mathbb{N}$ is a given natural number, then the m -tail of $(x_n)_{n \in \mathbb{N}}$

$$(x_{m+n})_{n \in \mathbb{N}} = (x_{m+1}, x_{m+2}, \dots)$$

Theorem 1: A sequence in $\mathbb{R}$ can have at most one limit.

Proof: Suppose x' and x'' are both limits of (x_n) . We shall show $x' = x''$

convince yourself that if we show that $\forall \epsilon > 0, |x' - x''| < \epsilon$ then we have proved the result.

(Observe that if $a \in \mathbb{R}$ and for every $\epsilon > 0, |a| < \epsilon$ then $a = 0$)

Thus we will show that $\forall \epsilon > 0, |x' - x''| < \epsilon$

Let $\epsilon > 0$ be arbitrarily (but fixed) chosen.

Then $\exists k' \in \mathbb{N}$ such that $|x_n - x'| < \epsilon/2 \quad \forall n \geq k'$
and $\exists k'' \in \mathbb{N}$ such that $|x_n - x''| < \epsilon/2 \quad \forall n \geq k''$

Let $K = \max\{k', k''\}$

$$\begin{aligned}
 \text{Then } |x' - x''| &= |x' - x_k + x_k - x''| \\
 &\leq |x' - x_k| + |x_k - x''| \\
 &< \epsilon/2 + \epsilon/2 = \epsilon
 \end{aligned}$$

Since $\epsilon > 0$ is arbitrary, we conclude that
 $|x' - x''| = 0 \Rightarrow x' = x''$ (QED)

Example ①: Prove that $\lim_{n \rightarrow \infty} \left(\frac{1}{n}\right) = 0$

Intuitively it is clear but we will show it using the definition (ϵ)

Proof: Let $\epsilon > 0$ be given (arbitrarily chosen but fixed)

Then there exists a natural number K (which depends upon $\epsilon > 0$) such that $\frac{1}{K} < \epsilon$

Thus if $n \geq K$, then $\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{K} < \epsilon$

This implies that $\lim_{n \rightarrow \infty} \left(\frac{1}{n}\right) = 0$

Exercises: Prove that

$$(1) \quad \lim_{n \rightarrow \infty} \left(\frac{1}{n^2 + 1}\right) = 0$$

$$(2) \quad \lim_{n \rightarrow \infty} \left(\frac{3n+2}{n+1}\right) = 3$$

$$(3) \quad \lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$$

(4) If $0 < b < 1$, then $\lim_{n \rightarrow \infty} (b^n) = 0$

Definition: A sequence (x_n) is said to be Bounded if there exists a real number $M > 0$ such that $|x_n| \leq M \quad \forall n \in \mathbb{N}$

Theorem (2): A convergent sequence is bounded.

Proof: (Skipped)

• However the converse is not necessarily true.

Example: The sequence $(-1, 1, -1, 1, \dots)$ is bounded but not convergent.

Theorem (3): Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be convergent sequences of real numbers.

(1) If $\forall n \in \mathbb{N}, x_n \geq 0$ then $\lim_{n \rightarrow \infty} (x_n) \geq 0$

(2) If $\forall n \in \mathbb{N}, x_n \leq y_n$ then $\lim_{n \rightarrow \infty} (x_n) \leq \lim_{n \rightarrow \infty} (y_n)$

(3) If $a, b \in \mathbb{R}$ such that $\forall n \in \mathbb{N}, a \leq x_n \leq b$ then $a \leq \lim_{n \rightarrow \infty} (x_n) \leq b$

(4) If $\lim_{n \rightarrow \infty} (x_n) = x$ then $\lim_{n \rightarrow \infty} (|x_n|) = |x|$

Definition: Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of real numbers.

We say that

(i) $(x_n)_{n \in \mathbb{N}}$ is an increasing sequence

$$\Leftrightarrow x_1 \leq x_2 \leq x_3 \leq \dots \leq x_n \leq x_{n+1} \leq \dots$$

(ii) $(x_n)_{n \in \mathbb{N}}$ is a decreasing sequence

$$\Leftrightarrow x_1 \geq x_2 \geq \dots \geq x_n \geq x_{n+1} \geq \dots$$

(iii) $(x_n)_{n \in \mathbb{N}}$ is a monotonic sequence

$$\Leftrightarrow (x_n)_{n \in \mathbb{N}} \text{ is either increasing or decreasing}$$

Theorem ④:

A monotone sequence of real numbers is convergent if and only if it is bounded.

Further

(a) If (x_n) is a bounded increasing sequence then $\lim_{n \rightarrow \infty} (x_n) = \sup \{x_n : n \in \mathbb{N}\}$ (least upper bound)

(b) If (x_n) is a bounded decreasing sequence then $\lim_{n \rightarrow \infty} (x_n) = \inf \{x_n : n \in \mathbb{N}\}$ (greatest lower bound)

Remark: Note the two statements:

(x_n) is a sequence in $\mathbb{R}$ that converges to $x \in \mathbb{R}$

$\nleftrightarrow (x_n)$ is a convergent sequence in $\mathbb{R}$.

Example (3)

Define $e_n = \left(1 + \frac{1}{n}\right)^n \quad \forall n \in \mathbb{N}$

Show that $(e_n)_{n \in \mathbb{N}}$ is

(a) bounded sequence

(b) an increasing sequence

Note:

Hence $(e_n)_{n \in \mathbb{N}}$ is a convergent sequence

(Actually $(e_n)_{n \in \mathbb{N}}$ converges to $e \approx 2.718281\dots$)

Infinite Series

• An expression of the form

$$x_1 + x_2 + \dots + x_n + \dots$$

is called an infinite series.

Question: How do we make sense of an
infinite number of addition?

Definition: If $(x_n)_{n \in \mathbb{N}}$ is a sequence in $\mathbb{R}$ then the partial sums generated by $(x_n)_{n \in \mathbb{N}}$ is the sequence $(s_k)_{k \in \mathbb{N}}$ defined by

$$\begin{aligned}s_1 &= x_1 \\s_2 &= x_1 + x_2 \\s_3 &= x_1 + x_2 + x_3 \\s_4 &= x_1 + x_2 + x_3 + x_4 \\&\vdots \\s_k &= x_1 + x_2 + \dots + x_k \\&\vdots\end{aligned}$$

If $\lim_{k \rightarrow \infty} (s_k)$ exists (ie. $\exists s \in \mathbb{R}$ such that $s_k \rightarrow s$)

we say that the series $x_1 + x_2 + \dots$ (generated by (x_n)) is convergent and call this limit

the sum or the value of the series.

If this limit does not exist, we say that the series is divergent.

We use the symbol $\sum_{n=1}^{\infty} x_n$ (or simply, $\sum x_n$)

to denote both the infinite series generated by the sequence (x_n) and also to denote the value $\lim_{k \rightarrow \infty} (s_k)$, in case the limit exists.

Example (Geometric Series):

Let $r \in \mathbb{R}$, $r \neq 0$

consider the sequence

$$(1, r, r^2, \dots) = (r^n)_{n \in \mathbb{N} \cup \{0\}}$$

Next consider the series generated by this sequence

$$\sum_{n=0}^{\infty} r^n = 1 + r + r^2 + \dots + r^{n-1} + r^n + \dots$$

(This series is called the "Geometric Series")

claim: If $|r| < 1$ then $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$

Proof: For any $k \in \mathbb{N}$,

$$S_k = 1 + r + r^2 + \dots + r^{k-1}$$

$$\text{Then } r S_k = r + r^2 + \dots + r^{k-1} + r^k$$

$$\text{Then } S_k(1-r) = 1 - r^k \Rightarrow S_k = \frac{1-r^k}{1-r}$$

Since $|r| < 1$, clearly $r \neq 1$.

We want to show that $\lim_{k \rightarrow \infty} S_k = \frac{1}{1-r}$

(Thus we wish to show that
 $\forall \epsilon > 0, \exists M \in \mathbb{N}$ s.t. $k \geq M \Rightarrow \left| S_k - \frac{1}{1-r} \right| < \epsilon$)

$$\text{Now } S_k = \frac{1-r^k}{1-r} = \frac{1}{1-r} - \frac{r^k}{1-r}$$

$$\Rightarrow S_k - \frac{1}{1-r} = \frac{-r^k}{1-r}$$

$$\Rightarrow \left| S_k - \frac{1}{1-r} \right| = \left| \frac{r^k}{1-r} \right| \leq \frac{|r|^k}{|1-r|}$$

Since $|r|^k \rightarrow 0$ when $|r| < 1$,
it follows that the geometric series
converges to $\frac{1}{1-r}$ when $|r| < 1$.

Example ④: Considering the sequence $((-1)^n)_{n=0}^{\infty}$
prove that the series
 $\sum_{n=0}^{\infty} (-1)^n$ diverges.

Proof:

$$S_1 = 1$$

$$S_2 = 1 - 1 = 0$$

$$S_3 = 1 - 1 + 1 = 1$$

$$S_4 = 1 - 1 + 1 - 1 = 0$$

Thus

$$S_n = \begin{cases} 1 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

clearly (S_k) is divergent.
Therefore $\sum_{n=0}^{\infty} (-1)^n$ is divergent.

Theorem : (The nth term test):

If the series $\sum x_n$ converges then $\lim_{n \rightarrow \infty} (x_n) = 0$

Note: Using this test you can show that a certain series does not converge. However, you can't use it to show that a certain series converges.

Facts:

- (1) The geometric series $\sum_{n=0}^{\infty} r^n$ diverges if $|r| \geq 1$
- (2) The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.
- (3) The p-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.
- (4) The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges when $p > 1$
- (5) The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ diverges when $0 < p \leq 1$
- (6) The alternating harmonic series given by:
$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{(-1)^{n+1}}{n} + \dots$$

is convergent.

Comparison Tests:

Theorem (Basic Comparison test):

Let (x_n) and (y_n) be sequences in $\mathbb{R}$.

If $\exists K \in \mathbb{N}$ such that $0 \leq x_n \leq y_n \forall n \geq K$

then (a) $\sum y_n$ converges $\Rightarrow \sum x_n$ converges.

(b) $\sum x_n$ diverges $\Rightarrow \sum y_n$ diverges.

Theorem (Limit Comparison Test):

Suppose (x_n) and (y_n) are strictly positive sequences and suppose that the following limit exists in $\mathbb{R}$. $r = \lim_{n \rightarrow \infty} \left(\frac{x_n}{y_n} \right)$

(a) If $r \neq 0$ then

$\sum x_n$ is convergent $\Leftrightarrow \sum y_n$ is convergent.

(b) If $r = 0$ then

$\sum y_n$ is convergent $\Rightarrow \sum x_n$ is convergent.

Exercises :

(1) The series $\sum_{n=1}^{\infty} \frac{1}{n^2+n}$ is convergent.

(2) The series $\sum_{n=1}^{\infty} \frac{1}{n^2-n+1}$ is convergent.

(3) The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1}}$ is divergent.

(4) The series $\sum_{n=1}^{\infty} \frac{1}{n!}$ is convergent.

• To show that a convergent sequence of real numbers is bounded.

Proof: Assume that the sequence (x_n) of real numbers converges to $x \in \mathbb{R}$.

Then for every $\epsilon > 0$, $\exists k = k(\epsilon) \in \mathbb{N}$ such that

$$|x_n - x| < \epsilon \quad \forall n \geq k$$

choose $\epsilon = 1$ (you can choose any number you like)

Then $\exists k_1 \in \mathbb{N}$ such that $|x_n - x| < 1 \quad \forall n \geq k_1$

$$\Rightarrow -1 < x_n - x < 1 \quad \forall n \geq k_1$$

$$\Rightarrow -1 < x_n - x < 1 \quad \forall n \geq k_1$$

$$\Rightarrow -1+x < x_n < 1+x \quad \forall n \geq k_1 \quad \left(\text{So, } |x_n| < \max\{|1+x|, |-1+x|\} \right)$$

Let $M = \max\{|-1+x|, |1+x|, |x_1|, |x_2|, \dots, |x_{k_1-1}|\}$

Then $|x_n| \leq M \quad \forall n \in \mathbb{N}$ and so (x_n) is bounded. (Q.E.D.)